LAB 7: COLLABORATION DIAGRAM - MULTI-VENDOR E-COMMERCE PLATFORM

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Academic Year: 2026-27

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AIM

To draw the UML Collaboration (Communication) Diagram for the Multi-Vendor Checkout scenario, illustrating structural relationships between participating objects and numbered message paths.

THEORY

A Collaboration Diagram (termed Communication Diagram in UML 2) emphasizes structural organization (which objects are linked to which) rather than explicit time sequences. Messages are labeled with sequence numbers (1, 2, 3...) along static communication links to indicate execution order.

- **Object / Actor:** Participants in the interaction represented as styled boxes.
- **Link:** Solid lines indicating direct structural relationships enabling communication.
- **Message & Direction:** Arrows accompanied by sequence numbers and method signatures showing call direction.

MESSAGE SEQUENCE SPECIFICATION

No.	From → To	Message Call	Description
1	Customer → :CheckoutEngine	`submitOrder(cartId, addrId)`	Customer submits cart items for purchase.
2	:CheckoutEngine → :InventoryManager	`lockStock(items)`	Reserves stock items in PostgreSQL DB.
3	:InventoryManager → :CheckoutEngine	`stockOK`	Returns inventory reservation confirmation.
4	:CheckoutEngine → :PaymentGateway	`chargeCard(amount)`	Sends authorization request to payment SDK.
5	:PaymentGateway → :CheckoutEngine	`txSuccess(txRef)`	Returns payment transaction verification reference.
5.1	:CheckoutEngine → :OrderSplitter	`generateSubOrders(cartId)`	Splits multi-vendor items into individual sub-orders.
6	:CheckoutEngine → Customer	`confirmRef(parentOrderId)`	Displays order confirmation number to Customer.

COLLABORATION DIAGRAM

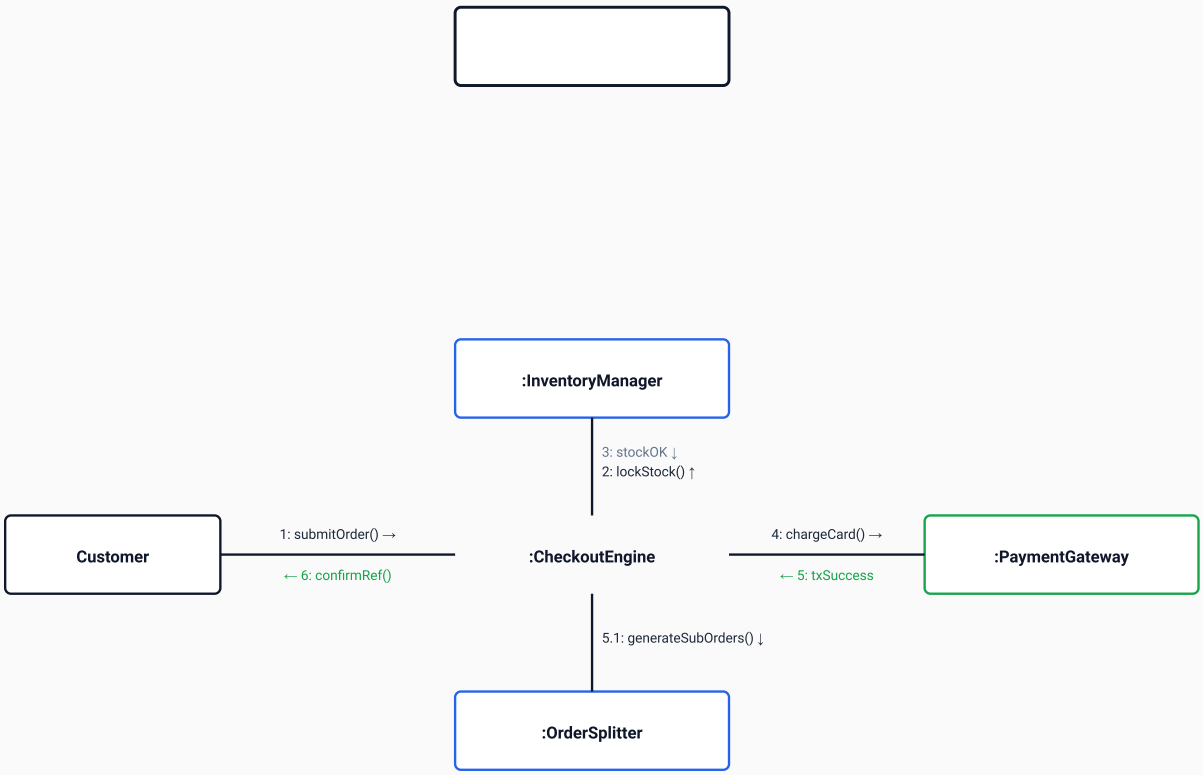


FIGURE 7.1: COLLABORATION DIAGRAM FOR MULTI-VENDOR ORDER PLACEMENT

CONCLUSION

The Collaboration Diagram was constructed, mapping communication links across CheckoutEngine, InventoryManager, PaymentGateway, and OrderSplitter objects, detailing structural message routing for multi-vendor checkout.